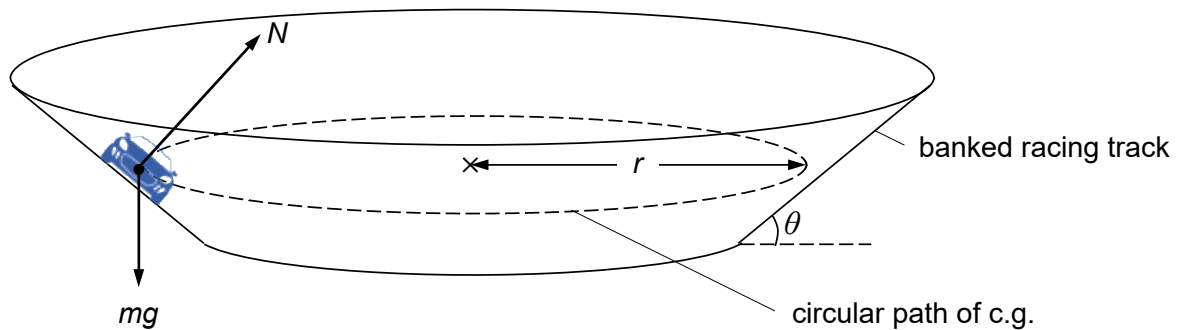


9. A racing car of mass m is moving with speed v on a banked racing track. The angle of inclination of the track is θ . The centre of gravity (c.g.) of the car moves along a circular path of horizontal radius r .



The centripetal force arises entirely from a component of the normal reaction N from the track.

Which of the following relations is correct?

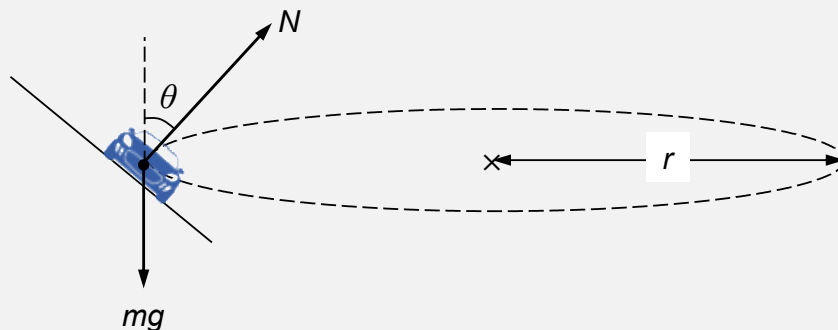
A $N \sin \theta = mg$

B $N = mg \cos \theta$

C $rg \tan \theta = v^2$

D $v^2 \tan \theta = rg$

Answer: C



horizontal direction: $N \sin \theta = \frac{mv^2}{r}$ (1)

vertical direction: $N \cos \theta = mg$ (2)

(1) / (2): $\tan \theta = \frac{v^2}{rg}$